Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/08/2023
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	01/01/2022	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per NSSC information (Ed., 12/29/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Carey, Van Ph.D. (University of California, Berkeley)		
Grant/Contract No.:	80NSSC22K0329		
Performance Goal No.:			
Performance Goal Text:			
	A program to study the flammability characteristics of wiring materials in spacecraft environments (that will complement the existing Japanese Space Agency (JAXA) project entitled, “Fundamental Research on International Standard of Fire Safety in Space”) is being carried out at U.C. Berkeley under NASA sponsorship, with the objective of providing information about electrical wires flammability in “Space Exploration Atmospheres” (SEA). The wires under study are similar to those of the tests being conducted at Hokkaido University by the Japanese counterpart researchers, and the tests are planned to be conducted in the International Space Station/Japanese Experiment Module (ISS/KIBO) unit. The long-term goal of the project is to understand the changes in electrical wire flammability, specifically the flame spread Limiting Oxygen Index - Maximum Oxygen Concentration (LOI-MOC), in the environment expected in		

Task Description:	space-based facilities – in particular, those at space exploration atmospheres (SEA). These environments include reduced gravity, low velocity flows, reduced pressures, and elevated oxygen concentrations. The near-term objective of the research is to conduct tests of wire burning in a ground-based Flammability Limits at Reduced-g Experiment (FLARE) UCB apparatus at varied flow velocities, reduced ambient pressure, and elevated oxygen concentration. Another objective of the project is to model the experimental data using Neural Networks to study the effectiveness of that approach in predicting wire burning behavior in spacecraft environments. The experiments and modeling support the ISS/KIBO experiments to be conducted under the FLARE/JAXA project. The project experimental methodology consists of obtaining flame spread rates at varied flow velocity and ambient pressure and oxygen concentration in wires with different cores and insulation materials (primarily those being studied at Hokkaido University and planned for the tests in the ISS/KIBO). The wires under study are primarily “laboratory” type wires consisting of a conducting core, primary copper, and a polymeric insulation (primarily PE). The UCB experiments are being conducted in normal gravity, with a limited number of tests conducted in an aircraft following a parabolic path. The testing methodology is supported with theoretical modeling using Neural Network and Genetic Algorithms to predict flame spread over wires at different geometrical and ambient conditions. This research primarily supports the FLARE/JAXA program but also could lead to better determination and ranking of the fire hazard characteristics of potential wire materials to be used in spacecraft for long-term exploration missions. NOTE this is continuation of "Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space," grant #80NSSC22K0329 with the same principal investigator, Prof. Carlos Fernandez-Pello. Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE).” JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in spacecraft allows us to accurately elucidate the effect of the environment parameters on the ignition and flame spread over combustible material, and through them their potential fire hazard. Particularly important is the determination of the Limiting Oxygen Concentration (LOC) on flame extinction under spacecraft environments. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Another important aspect of the research is the effect of melting and dripping of plastic insulation in normal gravity in comparison with microgravity. The results are relevant because dripping will not occur in microgravity and consequently could impact their burning and methodology to screen. The investigation and results have also benefits for terrestrial fire safety by providing further information about the flammability of materials under a variety of environments.
Task Progress:	A summary of relevant research progress during this reporting period (March 1, 2023-February 28, 2024) is presented here. The research consisted primarily in extending the modeling of the flame spread over insulated wires using Neural Networks (NN) to the combined use of NN and of Genetic Algorithms. A more detailed information about the research conducted can be found in the cited publications. Modeling Flame Spread Over Insulated Wires Using Neural Networks and Genetic Algorithms The objective of this theoretical task is to use artificial neural networks (ANN) together with genetic algorithms (GA) to further understand how certain variables in the problem of flame spread over insulated wires relate to one another and affect the insulation melting and dripping from the flame spread. The information will be used to model both flame spread rate and dripping of insulated electrical wires, and to develop predictive equations of the problem as a whole. The equations will be applied to predict burning behavior of wires in normal and microgravity environments. The research primarily supports the above-mentioned JAXA program but also could lead to a better determination and ranking of the fire hazard characteristics of potential wire materials to be used in spacecraft for long-term exploration missions. Here a summary of the work conducted during this reporting period is summarized. More detailed description of the work can be found in the references provided. 1. Comprehensive Database To create an artificial neural network that can predict flame spread rate along electrical wires through use of various input parameters, a comprehensive database of experimental flame spread rate results was compiled from the current existing work in the field. Altogether, the database consists of approximately 1200 data points, with approximately one third (400) coming from internal experiments and two thirds (800) coming from external sources. 2. Artificial Neural Network: Structure, Training and Validation To create a model of this comprehensive database, an ANN was trained to predict the flame spread rate along wires of different sizes and compositions under various ambient conditions. After training the ANN with half of the available data, validation of the ANN was carried out using the reserved other half of the data that was not put through the training process. Because the ANN has never seen this data before, it is able to validate the training of the ANN and show that it is actually able to make predictions and has not just memorized the patterns found in the training data. The results from the validation dataset are obtained by running the corresponding input parameters through the fully trained ANN for one epoch. 3. Parametric Trends While there is room for improvement in the ANN model, that is not to say that nothing further can be learned from it in its current state. Examining some of the parametric trends predicted by the ANN can give further insight into the flame spread rate along the electrical wire insulation problem. For example, parametric surfaces can be developed which demonstrate the ANN’s predictions for flame spread rate’s dependence on oxygen concentration and ambient pressure. The surfaces clearly display the well-known trend for flame spread rate to increase both with increasing oxygen concentration and increasing pressure; however, it also shows that, even at extremely low oxygen concentrations or pressures, a high value of the other parameter can still result in a moderately fast flame spread rate.

	3. Neural Network and Genetic Algorithm Using Non-Dimensional Parameters To improve the generality of the predictions and to assist with an equation formulation of the problem as a whole, an additional neural network model was trained using a selection of non-dimensional parameters. To determine the parameters to use in the model, a large set of possible non-dimensional combinations was generated and selected based on the optimal and lowest quantity of inputs. The selection was based on a set of generated neural network models, generated by a combination of nondimensional parameters. The highest precision was selected based on predictive/measured deviation by root mean square of the error for a given quantity of inputs. The selected non-dimensionalized neural network successfully maintained a low root mean square error of the predictive performance compared to the reported values. This performance was achieved with the more generalized non-dimensional input parameters, while also reducing the over-fitting by the large reduction in the fitting parameters as a result of the reduced number of input parameters.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Rivera J, San Martin D, Gollner M, Torres C, Fernadez-Pello C. "A machine learning approach to predict the critical heat flux of solid fuels." 14th International Symposium on Fire Safety Science, International Association of Fire Safety Science (IAFSS), Tsukuba, Japan, October 23-27, 2023. Abstracts. 14th International Symposium on Fire Safety Science, International Association of Fire Safety Science (IAFSS), Tsukuba, Japan, October 23-27, 2023. , Oct-2023
Abstracts for Journals and Proceedings	Scudiere C, Gagnon L, Fernandez-Pello C, Carey V. "Modeling analysis of experimental results of flame spread over insulated wires using an artificial neural network." The 9th International Workshop for ISS Material Flammability Standard Project: FLARE, Tsukuba, Japan, September 13-15, 2023. Abstracts. The 9th International Workshop for ISS Material Flammability Standard Project: FLARE, Tsukuba, Japan, September 13-15, 2023. , Sep-2023
Abstracts for Journals and Proceedings	Rivera J, San Martin D, Gollner M, Torres C, Fernadez-Pello C. "A machine learning approach to predict the critical heat flux of solid fuels." 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023. Abstracts. 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023. , Mar-2023